



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No : 9

TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

EXERCISE

1. Create a **Bank Account program** where account number is final and cannot be changed.
2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

CODE

```
exp9 > J bank.java > bank > display()
1  import java.util.Scanner;
2
3  class bank {
4      final int accountnumber;
5      String accountholder;
6      double balance;
7
8      // Constructor
9      bank(int accountnumber, String accountholder, double balance) {
10         this.accountnumber = accountnumber;
11         this.accountholder = accountholder;
12         this.balance = balance;
13     }
14
15     // Method to display account details
16     void display() {
17         System.out.println("Account Number : " + accountnumber);
18         System.out.println("Account Holder : " + accountholder);
19         System.out.println("Balance      : ₹" + balance);
20     }
21
22
23
24     Run main | Debug main
25     public static void main(String[] args) {
26         Scanner sc = new Scanner(System.in);
27
28         System.out.print("Enter Account Number: ");
29         int accountnumber = sc.nextInt();
30         sc.nextLine();
31
32         System.out.print("Enter Account Holder Name: ");
33         String accountholder = sc.nextLine();
34
35         System.out.print("Enter Initial Balance: ");
36         double balance = sc.nextDouble();
37
38         bank account = new bank(accountnumber, accountholder, balance);
39
40         account.display();
41     }
42 }
43
```

OUTPUT

```
● raghav_oze@raghavs-MacBook-Air-3 exp9 % java bank.java
Enter Account Number: 0001
Enter Account Holder Name: Agam Oze
Enter Initial Balance: 10000
Account Number : 1
Account Holder : Agam Oze
Balance      : ₹10000.0
✧ raghav_oze@raghavs-MacBook-Air-3 exp9 %
```

CODE

```
J Main.java  .gitattributes  J Student.java U  J bank.java U  J book.java U ●
exp9 > J book.java > book > main(String[] args)
1  import java.util.Scanner;
2
3  class book {
4      final String isbn;
5      String title;
6      String author;
7      double price;
8
9      book(String isbn, String title, String author, double price) {
10         this.isbn = isbn;
11         this.title = title;
12         this.author = author;
13         this.price = price;
14     }
15
16     void display() {
17         System.out.println();
18         System.out.println("ISBN   : " + isbn);
19         System.out.println("Title  : " + title);
20         System.out.println("Author : " + author);
21         System.out.println("Price  : ₹" + price);
22     }
23
24     Run main | Debug main
25     public static void main(String[] args) {
26         Scanner sc = new Scanner(System.in);
27
28         System.out.print("Enter ISBN: ");
29         String isbn = sc.nextLine();
30
31         System.out.print("Enter Book Title: ");
32         String title = sc.nextLine();
33
34         System.out.print("Enter Author Name: ");
35         String author = sc.nextLine();
36
37         System.out.print("Enter Book Price: ");
38         double price = sc.nextDouble();
39
40         book details = new book(isbn, title, author, price);
41         details.display();
42     }
}
```

OUTPUT

```
● raghav_oze@raghavs-MacBook-Air-3 exp9 % clear
● raghav_oze@raghavs-MacBook-Air-3 exp9 % java book.java
Enter ISBN: 0987654321
Enter Book Title: GOD is the best scriptwriter
Enter Author Name: Agam Oze
Enter Book Price: 150

ISBN    : 0987654321
Title   : GOD is the best scriptwriter
Author  : Agam Oze
Price   : ₹150.0
✧✧ raghav_oze@raghavs-MacBook-Air-3 exp9 %
```

Conclusion

We have successfully learnt and implemented the use of the final keyword in two programs